

Knowing When to Abstain: Calibration and Selective Prediction for Machine-Learning Network Intrusion Detection

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Abstract—Machine-learning intrusion detectors are routinely deployed with a single hard decision and no notion of when their output should be trusted. We show, on the public NSL-KDD benchmark, that three standard detectors (logistic regression, a random forest, and gradient boosting) are markedly over-confident: expected calibration error reaches 0.2138, and the strongest model still attains an error rate of 0.2028 on a test set whose attack mix it was trained on. We then study two remedies. First, post-hoc Platt scaling fitted on in-distribution data fails to repair calibration once the test distribution shifts, leaving error essentially unchanged. Second, selective prediction—allowing the detector to abstain on its least-confident inputs—reduces selective risk from 0.2028 to 0.1127 at 80% coverage, but only for models whose confidence is informative; logistic regression’s confidence is nearly useless for ranking errors (area under the risk–coverage curve 0.2355 versus 0.0576). Finally, using a controlled family-holdout protocol we quantify behaviour on *unknown* attacks: detection of a held-out family collapses (e.g. DoS from 0.8696 to 0.6147), and abstention rejects only part of these unknowns—and almost none when the attack mimics benign traffic (R2L). We conclude that selective prediction is a cheap, model-agnostic safety layer for ML intrusion detection, with clearly characterised limits.

Index Terms—intrusion detection, selective prediction, calibration, uncertainty, open-set recognition, NSL-KDD

I. INTRODUCTION

A network intrusion detector that emits only “attack” or “benign” gives an operator no way to tell a confident verdict from a guess. In practice this matters twice over: alerts compete for scarce analyst attention, and genuinely novel attacks—absent from any training corpus—are exactly the cases a static classifier is least equipped to label. A detector that could say “I am not sure” on these inputs, and defer them to a human or a secondary control, would be more useful than one forced to commit.

This paper asks a narrow, practical question: do off-the-shelf machine-learning intrusion detectors produce confidence scores that are trustworthy enough to drive such an abstention policy, and what do they do when faced with attacks they have never seen? We answer empirically on NSL-KDD [1], a public benchmark whose published test split deliberately contains attack types absent from training. Our contributions are:

- We measure the calibration of three standard detectors and show all are over-confident (Section IV-A), and that

post-hoc Platt scaling [2] fitted on in-distribution data does *not* transfer to the shifted test set.

- We evaluate selective prediction [3], [4] and show abstention cuts selective risk substantially, but that its usefulness is model-dependent: tree ensembles rank their errors well, a linear model does not (Section IV-B).
- We introduce a controlled family-holdout protocol for the *unknown-attack* regime and show detection collapses on held-out families while abstention provides only partial cover—and almost none for attacks that resemble benign traffic (Section IV-C).

Throughout we follow the evaluation guidance of Arp *et al.* [5]: the calibrator is fit only on held-out training data, and no test information leaks into model selection.

II. RELATED WORK

Calibration. Modern classifiers often output probabilities that do not match empirical accuracy [6], [7]. Expected calibration error (ECE) summarises this gap, and temperature/Platt scaling are common post-hoc fixes [2]. These analyses typically assume the evaluation data is drawn from the training distribution—an assumption that fails for intrusion detection.

Selective prediction. A classifier with a reject option [3] trades coverage for accuracy on what it does accept; the risk–coverage curve and its area [4] quantify the trade-off. Maximum softmax/posterior probability is a strong baseline confidence score [8].

ML for intrusion detection. The limits of learned detectors—especially on novel attacks and under distribution shift—were argued early [9] and remain a methodological hazard [5]. We do not propose a new detector; we ask whether existing ones know when they are wrong.

III. METHODOLOGY

Data. NSL-KDD [1] provides 125973 training and 22544 test records with 41 features (three categorical) and a binary normal/attack label; each attack maps to one of four families (DoS, Probe, R2L, U2R). Its test split contains 17 attack *types* (3750 records) that never appear in training—a built-in open-set test. Numeric features are standardised and categorical features one-hot encoded; the encoder is fit on training data only.

TABLE I
ACCURACY, CALIBRATION ERROR (RAW / AFTER PLATT SCALING), AND
SELECTIVE-PREDICTION QUALITY ON THE NSL-KDD TEST SET. LOWER
ECE AND AURC ARE BETTER.

Model	Acc.	ECE _{raw}	ECE _{platt}	AURC	$r(0.8)$
Logistic reg.	0.7543	0.2138	0.2143	0.2355	0.1945
Random forest	0.7751	0.1574	0.1856	0.0592	0.1376
Hist. grad. boost.	0.7972	0.1812	0.1833	0.0576	0.1127

Models. We use scikit-learn [10] logistic regression, a random forest, and histogram gradient boosting—deliberately standard detectors, since our subject is their confidence, not their architecture.

Calibration. For each model we report ECE (15 equal-width bins of the predicted class probability) before and after Platt scaling, the latter fitted on a 20% held-out split of the training set.

Selective prediction. The confidence of a prediction is its maximum class probability. Sorting the test set by confidence and accepting the most-confident fraction c yields the selective risk $r(c)$; we report the area under $r(c)$ (AURC) and $r(0.8)$.

Open-set protocol. To probe genuinely unknown attacks under control, we remove one attack family entirely from training (keeping benign traffic and the other families), retrain, and measure detection and confidence on that family’s test records. We also report, at a global threshold giving 80% coverage, the fraction of unknown-family records that abstention rejects.

IV. RESULTS

A. Detectors are over-confident

Table I reports accuracy and calibration. All three models are over-confident: ECE ranges from 0.1574 (random forest) to 0.2138 (logistic regression). Platt scaling, fit on in-distribution held-out data, does not help on the shifted test set—ECE is essentially unchanged for gradient boosting (0.1812 to 0.1833) and slightly worse for the random forest (0.1574 to 0.1856). Calibrating to the source distribution does not certify probabilities once the deployed distribution moves, as Fig. 2 illustrates.

B. Abstention helps—if confidence is informative

Fig. 1 shows risk–coverage curves. For gradient boosting, abstaining on the least-confident 20% of inputs lowers selective risk from 0.2028 to 0.1127; the random forest behaves similarly (AURC 0.0592). Logistic regression is the cautionary case: its confidence barely ranks its errors (AURC 0.2355), so abstention buys little (0.2457 to 0.1945). The lesson is that selective prediction is only as good as the underlying confidence signal, which here depends strongly on the model family.

C. Unknown attacks: collapse and partial cover

On the built-in novel attacks, detection drops sharply: the random forest flags 0.7696 of *known* attacks but only 0.2773 of novel ones. The controlled family-holdout (Fig. 3) confirms

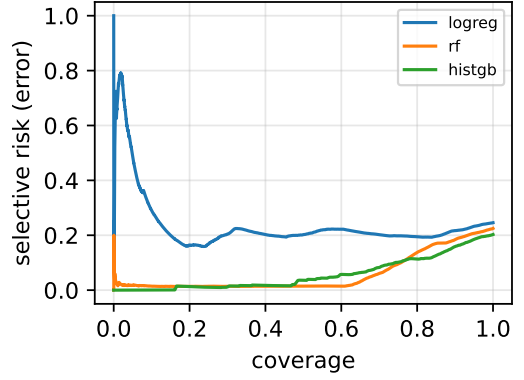


Fig. 1. Risk–coverage curves on NSL-KDD. Tree ensembles rank errors well; the linear model does not.

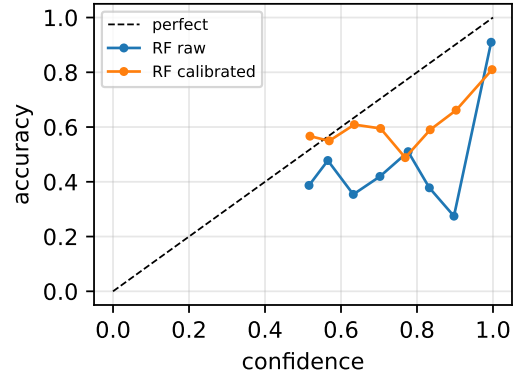


Fig. 2. Reliability of the random forest, raw vs. Platt-scaled, on the shifted test set.

this causally: detection of DoS falls from 0.8696 to 0.6147 when DoS is unknown, and Probe from 0.7918 to 0.4374. Abstention rejects 0.2918 of unknown DoS and 0.3172 of unknown Probe at 80% coverage—useful but partial. The R2L family is the sharp limit: detected at only 0.0586 even when *seen*, falling to 0.0035 when unknown, and abstention flags just 0.1737 of it. Because R2L traffic statistically resembles benign sessions, the detector is confidently wrong, and a confidence-based reject option cannot save it.

V. DISCUSSION AND LIMITATIONS

Selective prediction is a cheap, model-agnostic safety layer: at the cost of deferring a minority of inputs it materially lowers the error among those acted upon, and it partially flags unknown attacks. But our results bound the claim. First, the value of abstention is inherited from the confidence signal, which is good for tree ensembles and poor for the linear model. Second, post-hoc calibration on source data does not survive distribution shift, so calibrated probabilities should not be trusted as deployment-time guarantees. Third, and most important operationally, abstention only catches unknowns that *look* anomalous; attacks that mimic benign traffic (R2L here)

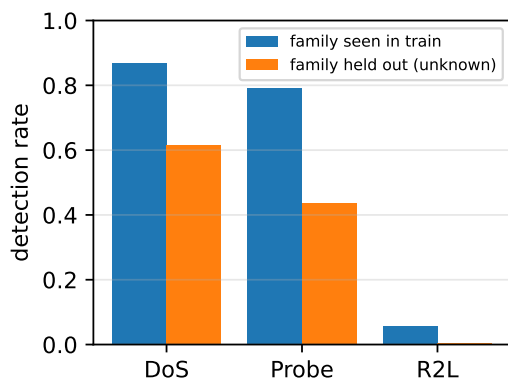


Fig. 3. Detection rate per family when the family is seen in training vs. held out (unknown).

are missed confidently and silently. NSL-KDD is dated and its feature set hand-crafted; confirming these effects on a modern flow-based corpus is the natural next step.

VI. CONCLUSION

On a standard benchmark, off-the-shelf intrusion detectors are over-confident, post-hoc calibration does not transfer under shift, and a simple reject option lowers selective risk and partially flags unknown attacks—except those that resemble normal traffic. Selective prediction is worth adding to ML intrusion detection, provided its confidence signal and its blind spots are measured rather than assumed.

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